

A Compensated Green-Tail Counterfamily for the Bateman–Turok Euclidean Torus

GPT-5.6.sol*

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Working draft, 17 August 2026

Abstract

The positive Euclidean lattice proposed by Bateman and Turok invites a deterministic route to reconstruction: prove that the complete action gradient cannot become softer than the free bilaplacian scale. We formulate that route as a scaled Polyak–Łojasiewicz inequality on the isotropic four-torus and disprove it. An explicit positive, nonseparable, polynomial-contrast family has action bounded above and away from zero while its normalized gradient quotient satisfies $Q_n/\omega_{L_n}^2 = O(n^{-2}) \rightarrow 0$. The construction uses an exactly mean-compensated critical source, the periodic graph Green operator, and a small four-way tensor perturbation. It realizes, rather than evades, the previously proved necessities of residual escape, unweighted-curvature flatness, height-cut current cancellation, and nonvanishing action. The result refutes one all-field deterministic proof architecture only. It does not decide the Witten quotient, Gibbs typicality, the interacting H^{-1} moment, continuum reconstruction, Born/Krein interpretation, or any Lorentzian claim.

1 Question and exact boundary

Bateman and Turok propose to control a higher-derivative Euclidean theory by a positive lattice weight and a hidden ghost-parity structure [1]. Before reflection positivity, continuum limits, or operator reconstruction can be addressed, one may ask a more elementary finite-volume question: is there a deterministic action-gradient floor at the free spectral scale?

Let $\mathbb{T}_L^4 = (\mathbb{Z}/L\mathbb{Z})^4$, with nearest-neighbor graph Laplacian Δ , and let $u : \mathbb{T}_L^4 \rightarrow (0, \infty)$. Define

$$r_x = \frac{\Delta u_x}{u_x}, \quad A(u) = \frac{1}{2} \sum_x r_x^2, \quad g = \nabla_{\log u} A, \quad Q(u) = \frac{\|g\|_2^2}{\|r\|_2^2}.$$

The free gap is

$$\omega_L = 4 \sin^2\left(\frac{\pi}{L}\right).$$

The proposed deterministic route would require a volume-independent $c > 0$ such that $Q(u) \geq c\omega_L^2$ for every nonconstant positive field. Our result is the negation of precisely this statement.

Theorem 1.1 (Compensated Green-tail counterfamily). *There are explicit integers $L_n \rightarrow \infty$ and positive nonseparable fields u_n on $\mathbb{T}_{L_n}^4$ for which constants $0 < c < C < \infty$, independent of n , satisfy*

$$c \leq \|r(u_n)\|_2^2 \leq C, \quad \frac{\max u_n}{\min u_n} = O(L_n^{5/12}), \quad \frac{Q(u_n)}{\omega_{L_n}^2} = O(n^{-2}) \rightarrow 0.$$

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Consequently no positive volume-independent constant gives the all-field scaled inequality above.

The dependency tags of Theorem 1.1 are exactly **LOCAL-ALGEBRAIC** and **EUCLIDEAN-SPECTRAL**. It is neither a **REDUCED-MODE** nor a **LORENTZIAN-CAUSAL** theorem.

2 The weighted-current identity

The useful variable is not r by itself but

$$h_x = \frac{r_x}{u_x^2}, \quad c_{xy} = u_x u_y.$$

Direct differentiation, retaining both orientations of every edge, gives the exact identity

$$g_x = \sum_{y \sim x} c_{xy} (h_y - h_x) =: (L_c h)_x. \quad (1)$$

Thus the nonlinear action gradient is a weighted graph Laplacian applied to an unweighted curvature. This identity explains both the concentration theorems and the counterfamily: $r = u^2 h$ can retain finite energy at large field height while h becomes flat and the weighted currents cancel.

For the height set $S_K = \{x : u_x > K\}$, internal currents cancel and

$$\Gamma_K := \sum_{x \in S_K} g_x$$

is exactly the net current across its boundary. Pairing with the centered cut indicator and using $\omega_L \geq 16/L^2$ gives

$$\frac{Q}{\omega_L^2} \geq \frac{\Gamma_K^2}{4\pi^4 \|r\|_2^2}. \quad (2)$$

The counterfamily must therefore cancel every fixed-height net current; it cannot be a single uncanceled spike.

3 The construction

Take the integer scales

$$L_n = n^{24}, \quad \lambda_n = n^{16}, \quad R_n = n^{21}, \quad \epsilon_n = \frac{\lambda_n}{R_n^2} = n^{-26}.$$

Let $d_L(x, 0)$ denote periodic graph distance and set

$$f_n(x) = \frac{8\lambda_n^3}{(\lambda_n^2 + d_{L_n}(x, 0)^2)^3}. \quad (3)$$

There is a unique mean-zero solution

$$-\Delta v_n = f_n - \bar{f}_n. \quad (4)$$

Define

$$u_n^0 = v_n - \min v_n + \epsilon_n.$$

The subtraction of $\bar{f}_n$ is load-bearing. It distributes the compensating negative source through the periodic Green tail; a local cutoff would create an order-one shell and restore a large gradient.

To exclude any hidden product ansatz, put

$$p_n(x) = \prod_{a=1}^4 \sin\left(\frac{2\pi x_a}{L_n}\right), \quad u_n = u_n^0(1 + n^{-1}p_n).$$

The factor is positive. Its four-way Fourier coefficient has nonzero pairing with $\log u_n$ and is orthogonal to every sum of one-coordinate functions; hence u_n is genuinely nonseparable.

4 Estimates

The proof uses exact finite-difference inequalities and four uniform analytic estimates. The detailed rational witnesses and the independent finite fixture are stored in the certificate named in §8.

Lemma 4.1 (Action does not vanish). *There are dimension-only $c, C > 0$ such that*

$$c \leq \left\| \frac{\Delta u_n^0}{u_n^0} \right\|_2^2 \leq C.$$

The inner annuli of the critical source supply the lower bound. Periodic Green estimates, split dyadically at λ_n and R_n , control the tail and give the upper bound. The lift ϵ_n prevents a zero of the denominator without erasing the inner contribution.

Lemma 4.2 (Gradient estimate). *The base field satisfies*

$$\|g(u_n^0)\|_2^2 \leq C \left(\lambda_n^{-8} + \frac{\lambda_n^4}{R_n^8} + \frac{R_n^4}{L_n^8} + \frac{\lambda_n^4}{R_n^4 L_n^4} \right).$$

The four terms respectively record the smoothed core, finite-radius tail, periodic compensation, and core–tail interaction. They are kept separate in the certificate so that no asymptotic cancellation is assumed.

Lemma 4.3 (Stable nonseparable perturbation).

$$\|r(u_n) - r(u_n^0)\|_2 = O(n^{-1}), \quad \|g(u_n) - g(u_n^0)\|_2 = O(n^{-1}L_n^{-2}).$$

Since $\omega_L^2 \geq 256/L^4$, substitution of the four scales yields

$$\frac{Q(u_n)}{\omega_{L_n}^2} \leq C(n^{-32} + n^{-8} + n^{-12} + n^{-20} + n^{-2}).$$

This proves Theorem 1.1. Notice that the slowest term is the deliberate nonseparability perturbation; the radial Green-tail base collapses faster.

5 How the proof search narrowed the geometry

The final construction was not guessed in isolation. A sequence of exact results removed simple architectures and derived necessary properties of any counterfamily.

Stage	What was established	What remained
RF-83–85	Cycle hierarchies collapse only relative to a diameter method; one-phase torus pullbacks and independent tensor phases move away from the free scale.	A genuinely coupled four-dimensional corrector.
RF-86–92	Sparse-flow, top-band, dyadic and virial bounds progressively exclude super-polynomial contrast and action density above $272/29$.	A bounded-action, polynomial-contrast sequence.
RF-93–94	Collapse at positive action forces residual energy above every fixed height, flat $h = r/u^2$, and cancellation across every height cut.	Compatibility of all three conditions.
RF-95	A uniform small-action floor excludes $A \rightarrow 0$.	The positive-action concentration branch.
RF-96	The compensated Green tail realizes that branch and refutes the all-field floor.	Probabilistic, interacting, continuum and reconstruction questions.

This history matters logically. RF-93–95 are not failed attempts at the final theorem: they are necessary-condition theorems that the counterfamily must and does satisfy. Conversely, RF-96 does not invalidate the earlier large-action or large-contrast lower bounds; it lives in their surviving region.

6 What the counterfamily changes

The theorem removes a tempting shortcut. Uniform deterministic coercivity of the complete finite-volume score cannot by itself carry the free bilaplacian gap through all positive field configurations. Any successful reconstruction argument must use information absent from that statement. Three possibilities remain conceptually distinct:

1. a probabilistic theorem showing that Green-tail neighborhoods have negligible Gibbs weight;
2. an interacting H^{-1} or Witten-form estimate whose negative or nonlocal terms see more than $\|\nabla A\|^2 / \|r\|^2$;
3. a reconstruction mechanism that needs only typical or observable sectors rather than an all-field deterministic floor.

The first route requires an entropy-versus-action comparison for a positive radius tube around the family, not just the weight of one configuration. The second requires the actual interacting operator, not an analogy with the free gap. The third must still establish reflection positivity or its stated replacement, a probability rule, and a controlled continuum limit.

7 Claim boundary

The theorem does *not* establish any of the following:

- failure of the full Bateman–Turok proposal or of its Witten quotient;
- non-negligible Gibbs probability of the counterfamily;
- failure of an interacting H^{-1} moment or spectral gap;
- a continuum counterexample or obstruction to Osterwalder–Schrader reconstruction;
- a Born rule, Krein probability interpretation, positive physical state, particle space, scattering or unitarity;
- any claim about Lorentzian propagation, BV Green operators, Hadamard states, renormalized products or a Lorentzian QME.

These exclusions are not cautionary prose appended after the fact. They are machine-readable negative claim flags in the certificate and claim map.

8 Certification and reproduction

The authoritative result is

```
reverse_physics/certificates/REVERSE_PHYSICS_BT_EUCLIDEAN_TORUS_GREEN_TAIL_COUNT
ERFAMILY_V1.json.
```

It records the scale ledger, exact inequalities, dependency tags, controls, input hashes, assumptions and exclusions. Its verifier is independent of the exploratory floating-point scout. An exact rational 4^4 -torus fixture reduces the source-centered geometry to 15 symmetry orbits while retaining all 256 vertices; it checks the Green solve, positivity, residual, weighted-current identity, nonseparability witness and exponent ledger.

The affected deterministic test chain is run by

```
python3 -m unittest discover -s reverse_physics/tests \
-p 'test_bt_euclidean_torus*.py'
python3 paper/generate_22_bateman_turok_torus_claim_map.py --check
python3 paper/verify_22_bateman_turok_torus_claim_map.py
```

The paper claim map hash-pins RF–83 through RF–96. Re-running a producer is reproduction; the independent verifier is the separate audit rail.

References

- [1] S. Bateman and N. Turok, “Escape from Ostrogradsky via hidden ghost parity,” 2026. <https://arxiv.org/abs/2607.00096>.
- [2] OpenAI Codex and A. A. Palm, “Reverse foundations of physics: separating physical postulates, mathematical carriers, and logical strength,” Paper 21 of this programme, 2026; `paper/21-reverse-foundations-of-physics.pdf`.